

Week 1 Office Hour Notes

1.0 Errata

Although I explain general Mathematical Induction in these notes, it's really there for review and honestly obsolete as far as week 1 homework (and homework in the future) goes. I incorrectly thought that students couldn't use structural induction due to the fact that there isn't a section on "induction and recursion" until week 4. Apologies for the confusion that this caused.

1.1 Homework 1 Question 3 Hints

We want to prove the result by induction on the length n of symbols in the wff

In particular, we need to use strong induction, we need to use the inductive hypothesis in the inductive step (i.e., assuming that α has length $n + 1$) on wff's that have $\leq n$ length (including instances where you have $< n$ (STRICTLY LESS) length than n , where you would need strong induction)

1.2 Mathematical Induction Review

We will wait until week 3 to fully derive these principles, but for now, I'll state the "principle of mathematical induction" (less commonly called the "principle of *weak* mathematical induction") in contrast to the "principle of strong mathematical induction".

Fact 1.1. (Principle of (weak) mathematical induction) To show that a result works for all natural numbers $\mathbb{N} := \{1, 2, \dots\}$ (or what we call an "inductive set", i.e. a set that we can index by the natural numbers and has a notion of a "successor property", which we'll more rigorously talk about come week 3), it suffices to first show the following:

1. Base Case. The result works for the $n = 1$.

2. Inductive Step. Assuming that the result works for some arbitrary $n \geq 1$, show that the result also works for $n + 1$.

In other words, let φ be some property/relation on $\mathbb{N}$. Then the above principle states that $\varphi(n)$ holds for all $n \in \mathbb{N}$ iff $\varphi(1)$ holds and we show that the following property

$$\forall (n \in \mathbb{N})(\varphi(n) \implies \varphi(n + 1)),$$

holds. Note that the assumption that $\varphi(n)$ holds for some $n \in \mathbb{N}$ we call the **inductive hypothesis**.

Example 1.1. For an easy example to apply induction, that is arithmetic-related, and not logic-related, let's show that $n^2 \geq n$, for all $n \in \mathbb{N}$ by induction on n . Here is what we do for the base case and the inductive step, respectively

Base Case: Note that for $n = 1$ in the base case, we have

$$n^2 = 1^2 = 1 \geq 1 = n,$$

and we are done.

Inductive Step: Assume $n^2 \geq n$ and we want to show $(n + 1)^2 \geq n + 1$. First, observe that since $2n \geq 0$ (since 0 as it's defined is usually defined to be a number smaller than the positive natural numbers), it follows by the *addition property of inequalities* that

$$2n + (n^2 + 1) \geq 0 + (n^2 + 1),$$

hence

$$(n + 1)^2 = n^2 + 2n + 1 = 2n + (n^2 + 1) \geq 0 + (n^2 + 1) = n^2 + 1 \implies (n + 1)^2 \geq n^2 + 1. \quad (1.1.1)$$

Since by *addition property of inequalities* applied to the inductive hypothesis $n^2 \geq n$, we have

$$n^2 + 1 \geq n + 1,$$

we conclude by (1.1.1) that

$$(n + 1)^2 \geq n^2 + 1 \geq n + 1.$$

To proceed with the next example, note that the nonnegative integers, i.e. $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$, is also an inductive set, with the "base case" $n = 0$ instead of $n = 1$.

Example 1.2. On one of the **Enderton 1.1** examples (which we discussed on the week 1 forum), we wanted to show that any wff α has the same number of right parentheses as it does left parentheses. Let's slightly change this example and first prove this for wff's containing only possibly the negation connective (or no connective at all). To go about this,

recall the following fact about the definition of wff's:

Fact 1.2. wff's are either "atomic" sentence symbols A_i OR they are of the form $(\neg\alpha), (\alpha \vee \beta), (\alpha \wedge \beta), (\alpha \rightarrow \beta), (\alpha \leftrightarrow \beta)$, where α, β are themselves wff's. So you see that any time that a wff is NOT just a sentence symbol A_i , any other case must begin and end with a left and right parenthesis respectively.

We can show this for all formulas by showing it works for any such formula α of any number $k \geq 0$ of left parentheses, by induction on the number of left parentheses $k \geq 0$ of a formula.

Base case: It could be that $k = 0$, and in that case, we must have a formula that is a single sentence symbol A_i , or else, if we have at least one right parenthesis, then a wff by definition (see the above fact) must begin at a left parenthesis and end at a right parenthesis, however beginning at a left parenthesis contradicts the hypothesis that $k = 0$.

We are then done vacuously since there's no parenthesis in general, and so the number of parentheses on both sides are the same.

Inductive Step: Assume that α has $k + 1$ right parentheses and assume our result holds for $k \geq 0$. Note that α cannot be simply a sentence symbol A_i since otherwise α would have 0 parentheses and not $k + 1$ parentheses.

The conclusion then more or less follows from the argument I gave in reply to the week 1 forum topic started by Daniel Phang August 28 1:55pm, since it follows by the condition that α only contains the negation symbol, that we are left with the case where α is of the form $(\neg\beta)$ where

AG Alex Goodlad Teacher
Edited Aug 30 12:11pm

I take it that this is the case of the inductive step where if α has $k + 1$ left parentheses and is $(\neg\beta)$, for some wff β . Let k_β be the number of left parentheses of β . Noting that $k_\beta + 1 = k + 1 \implies k_\beta = k$, we find that β has k left parentheses, triggering the inductive hypothesis, and it follows that β has k right parentheses. Therefore, α , as $(\neg\beta)$, has one more right parenthesis than β , so we conclude that α has $k + 1$ right parentheses, which is the same number of right parentheses that α has.

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To proceed with induction for many of the exercises in **Enderton 1.1**, we actually need the

Principle of Strong Mathematical Induction. To see this, let's look at the following example that also comes from that section of Enderton.

Example 1.3. On one of the **Enderton 1.1** exercises (which we discussed on the week 1 forum), we wanted to show that any wff α has the same number of right parentheses as it does left parentheses, and unlike in **Example 1.2**, this time we don't apply any restrictions to its connectives.

As in the previous example, We can show this for all formulas by showing it works for any such formula α of any number $k \geq 0$ of left parentheses, by induction on the number of left parentheses $k \geq 0$ of a formula. Note that the base case is exactly the same as the base case in that previous example. And for the inductive step, if we assume the result holds for $k \geq 0$ and α has $k + 1$ parentheses, as shown with the previous example, α cannot be a sentence symbol A_i , so we

are left with the cases where α is $(\neg\alpha_1)$, $(\alpha_1 \vee \alpha_2)$, $(\alpha_1 \wedge \alpha_2)$, $(\alpha_1 \rightarrow \alpha_2)$, or $(\alpha_1 \leftrightarrow \alpha_2)$, where α_1, α_2 are wff's, by definition.

Note that the case where α is $(\neg\alpha_1)$ works exactly the same as in the inductive step in the previous example and using weak induction still works. However, we get into trouble if we assume α is any of $(\alpha_1 \vee \alpha_2)$, $(\alpha_1 \wedge \alpha_2)$, $(\alpha_1 \rightarrow \alpha_2)$, or $(\alpha_1 \leftrightarrow \alpha_2)$, since in any of those other cases, we find that the number of left parentheses for α made up of the opening left parenthesis, the left parentheses of α_1 , and the left parentheses of α_2 , and hence we have

$$k + 1 = k_1 + k_2 + 1 \implies k_1, k_2 \leq k,$$

where $k_1, k_2 \geq 0$ are the number of left parentheses of α_1, α_2 , respectively. Since it's not necessarily true that either k_1 or k_2 is equal to k exactly, we cannot necessarily use the weak inductive hypothesis to state that α_1 and α_2 have the same number of right and left parentheses (which from there would pretty straightforwardly lead to our conclusion following). If there was a way we can assume that our hypothesis holds for not just $k \geq 0$ but for any $0 \leq k' \leq k$, then our problem is solved.

Turns out the principle of Strong mathematical Induction does just that!

Fact 1.3. (Principle of strong mathematical induction) To show that a result works for all natural numbers $\mathbb{N} := \{1, 2, \dots\}$ (or an "inductive set", as described previously), it suffices to first show the following:

1. Base Case. The result works for the base case $n = 1$.

2. Inductive Step. Assuming that the result works for all $1 \leq k \leq n$ some arbitrary $n \geq 1$, show that the result also works for $n + 1$.

Note that one can derive the principle of strong mathematical induction from the principle of weak mathematical induction, and usually, both principles of mathematical induction are derived from defining a so-called "inductive set". Additionally, we have what's called "structural induction" where we can prove results about wff's by induction by taking sentence symbols as the "base case", and then for the "inductive step", looking at α as

$(\neg\alpha_1), (\alpha_1 \vee \alpha_2), (\alpha_1 \wedge \alpha_2), (\alpha_1 \rightarrow \alpha_2)$, or $(\alpha_1 \leftrightarrow \alpha_2)$, where α_1, α_2 are formulas where we can assume the hypothesis holds. This concept, along with discussing the connections with induction, will not be discussed until week 3, and you will be expected to prove results only by regular induction (using either the strong or weak principles) and not structural induction.

EDIT: Although structural induction is derived in week 4, we actually do go over an overview of structural induction this week and can definitely use structural induction in the proofs for homework 1.